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Polycentric Urban Regions and the Quest for Synergy: Is a Network of Cities More than the Sum of the Parts?

Evert Meijers

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Summary. Polycentric urban regions, or urban networks, are often associated with the notion of synergy, the assumption being that the individual cities in these collections of distinct but proximally located cities relate to each other in a synergetic way, making the whole network of cities more than the sum of its parts. Drawing on economic network theories, an analysis of the presence of synergy is carried out for the Randstad region in the Netherlands, which is often considered a classic example of a polycentric urban region. The analysis focuses on the synergy mechanisms of co-operation and in particular complementarity. The results are mixed. In terms of co-operation, the Randstad has become more synergetic. However, the less complementary economic roles of the cities caused a reverse effect.

1. Introduction

Interest in polycentric urban regions has increased rapidly during the past decade. This can be partly explained by the emerging belief that regions in general are becoming the most important spatial level of international territorial competition. In addition, polycentric urban regions are believed to be the next stage in the expansion of urban living space, particularly in densely populated countries or regions. The daily urban space of actors would not only cover the city, its suburbs and its surrounding rural area, but would extend to include other cities as well. In polycentric urban regions, cities seem to have coalesced in functional and morphological terms into larger and more dispersed regional urban systems. In the literature, polycentric urban regions are often defined as

collections of historically distinct and both administratively and politically independent cities located in close proximity and well connected through infrastructure (see Kloosterman and Lambregts, 2001). However, as the literature on polycentric urban regions is still limited and therefore rather unconsolidated (Bailey and Turok, 2001), a diversity of concepts is applied, which are largely synonymous with the polycentric urban region concept used here. Recent examples include ‘multicore city-regions’ (Westin and Östhol, 1994), ‘network cities’ (Batten, 1995), ‘city networks’¹ (Camagni and Salone, 1993) and ‘polynucleated metropolitan regions’ (Dieleman and Faludi, 1998a). Moreover, in terms of ideas on spatial structure and interurban relationships, the polycentric urban region concept builds on older concepts such as the ‘dispersed city’ (Burton, 1963),

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'megapolis' (Gottmann, 1961) or the ideas of Stein and Mumford on 'the regional city' (Stein, 1964).

Interest in polycentric urban regions has also increased among politicians and urban planners, as evidenced by their appearance as planning concepts in planning policies in a wide variety of European countries including Belgium ('urban networks'), Denmark (polycentric 'national centres'), Estonia ('urban networks'), France ('*réseaux des villes*'), Germany ('*Metropolregionen*'; '*Städtenetze*'), Greece ('twin poles'), Italy ('city networks'), Ireland ('linked gateways'), Lithuania ('Metropolis Vilnius–Kaunas'), the Netherlands ('urban networks'), Poland ('*duopols*') and Switzerland ('*vernetzte Städtessystem*'). Interestingly, planning policy concepts referring to polycentric urban regions often make use of the network metaphor. This is by no means a coincidence. The metaphor of the network emphasises the complex and strong relationships between the cities and thus the coherence and unity of the region. Moreover, networks are associated with economies of scale, critical mass and synergy. Not surprisingly, the network metaphor has become part of the standard vocabulary of administrators, planners and other policy-makers promoting or otherwise dealing with polycentric urban regions. In particular, the idea of synergy, or 'being more than the sum of the parts' is a central objective in many policies for polycentric urban regions. Examples are manifold, but can for instance be found in the Flanders Structural Outline (Ministerie van de Vlaamse Gemeenschap, 1997) which introduces the concept of the Flemish Diamond (see also Albrechts, 1998) and in the proposals for a new national spatial policy in the Netherlands in which the concept of 'urban networks' (for instance, the Randstad region) plays a prominent role (Ministerie van Volkshuisvesting, Ruimtelijke Ordening en Milieubeheer, 2004). The perhaps rather loose collection of cities in both regions would suddenly turn into a metropolis, and planners have suggested that the Randstad and Flemish Diamond could compete with the highest-level metropolitan agglomerations such as London and Paris, claiming that they provide

economies of scale without incurring the costs or agglomeration diseconomies that these large metropolises entail.

This article addresses the more theoretical underpinning of this quest for synergy by policy-makers. Its aim is to explore whether there is a theoretical basis for assuming the presence of synergy in polycentric urban regions and, building on that, to develop a conceptual model to analyse the presence of synergy in polycentric urban regions. Subsequently, the presence of synergy in the Randstad region is explored. The structure of this article is as follows. First, an overview of economic network theories in relation to the principle of synergy is presented. Assuming that polycentric urban regions may be considered as a network in some sense, this may shed some light on how synergy within such regions can be achieved as the same network fundamentals and synergy principles are likely to apply. The third section addresses this issue of synergy in polycentric urban regions, resulting in a conceptual model to analyse the presence of synergy. Before applying this model to the Randstad in sections 5 and 6 in order to establish whether there is synergy developing, the fourth section introduces this 'prime' or 'classic' example of a polycentric urban region (Hohenberg and Lees, 1985; Batten, 1995).

2. Synergy in Networks

The concept of networks is strongly linked to the notion of synergy. The aim of this section is to present some fundamental knowledge on network synergies, drawing on recent analysis of synergy in economic theory. The networks considered in these analyses are mainly networks between firms, transport networks and communication networks.

The word 'synergy' comes from the Greek (syn + ergos) and refers to a 'situation in which the effect of two or more co-operating or combined bodies or functions is larger than the sum of the effects each body or function alone can achieve. Synergy is often popularly formulated as $1 + 1 > 2$, which,

however, emphasises well that synergy can be expressed as the rise in performance of a network through efficient and effective interaction. Although widely used in economic theory, synergy is a rather fuzzy concept as it hides different mechanisms from which synergy is derived. Capello and Rietveld, analysing the synergy concept in a variety of economic theories at the micro-, meso- and macro-levels arrive at three distinct meanings

- Synergy means that when two or more actors co-operate, there is a positive result for both of them;
- synergy means that when co-operative behaviour is implemented in order to exploit complementarity in the production of a particular good, advantages are obtained by economic actors taking place in a group;
- synergy means that when individuals or firms ‘voluntarily and non-voluntarily’ are part of a group, externalities may be present and exploited by these individuals or groups in conducting their economic activity (Capello and Rietveld, 1998, p. 64).

The presence of one of these three synergy-releasing mechanisms—co-operation, complementarity or externality—combined with network behaviour generates synergy and hence economic benefits for the actors.

Next to these different mechanisms, the type of network is also relevant. Different synergy mechanisms play a role in the various types of network. Networks in general are made up of nodes (cities, households, firms, organisations, individuals), linkages between the nodes (infrastructure, relationships, ties), flows (people, goods, information, capital) and meshes.

Networks can be labelled as being of the ‘club’ type or of the ‘web’ type (Capineri and Kamann, 1998). In club networks, actors share a common objective, activity or service, while also having parallel interests and transaction chains. A classic but telling example is the tennis club. Members of such a club cannot afford the common objective—facilities for tennis—on their own, but by organising themselves they can. More members lead to positive externalities such as

lower membership fees or extended opening hours. Negative externalities arise, however, when all members want to play tennis at the same time. Web networks, on the other hand, are characterised by different activities of the actors. These are complementary instead of similar and are linked in a serial way. A typical example is a chain of enterprises or business units each undertaking a certain phase in the production of a product.

Distinguishing between club and web networks is relevant as synergy in both networks is achieved in different ways. In club networks, so-called ‘horizontal synergy’ can be achieved; in web networks, ‘vertical synergy’. In the case of horizontal synergy, the synergy derives from co-operation leading to economies of scale and so-called positive network externalities. Characteristic of these economies of scale is that they apply only to participants in the network. Externalities are present when the costs of participating in the network are less than the benefits of the co-operation. The value of co-operation is well articulated by Capineri and Kamann

Actors have the choice between independent ‘stand alone’ strategies where they perform all activities themselves usually at higher costs resulting in lower performance and strategies of co-operation which result in the transfer of activities and/or resources to other actors increasing a large range of types of dependencies but also improving their performance (Capineri and Kamann, 1998, p. 42).

Vertical synergy is the surplus value following from agglomeration or specialisation effects. Complementarity is the key synergy mechanism here. The synergy results from a specialisation process, redistributing resources and activities among the participating actors according to their competence. This means that the individual performance of actors improves as they can focus their efforts on their core activities, abandoning non-core activities unnecessarily absorbing energy (Capineri and Kamann, 1998). Classic network development theories also point to specialisation as the outcome of

a process of rationalisation of the network structure once it has achieved a considerable level of complexity.

To summarise, synergy is achieved through the mechanisms of co-operation, complementarity and externalities linked to them. Co-operation leads to horizontal synergy possibly achieved in club-type networks, complementarity to vertical synergy possibly achieved within web-type networks. Externalities are present in both. They represent the most important economic advantage of network behaviour

It is ... a matter of exploiting scale economies in complementary relationships and synergic effects in co-operative activities, achieved through participation in the network (Capello, 2000, p. 1927).

3. Synergy in Polycentric Urban Regions

Transferring the concept of synergy to cities does not seem too difficult. In fact, the first cities emerged because of synergy, developing from the advantages that arose from agglomeration economies. Living and working in cities entail advantages such as the supply of public services, specialised products and services, a large and diversified urban market and easy exchange and availability of information. The question, however, is how such agglomeration economies can also be organised in a network of cities.

This brief analysis of the synergy concept and some network fundamentals sheds some light on the way synergy can be achieved in polycentric urban regions. The cities making up a polycentric urban region can be considered the nodes in a network that is further made up by infrastructure, interurban relationships and flows. A city in itself is an accumulation of many other nodes such as households, firms, individuals, organisations, each one connected to other nodes by infrastructure, flows and interdependencies. So, in a polycentric urban region, a multitude of other networks can be found, by no means restricted to the scale of the polycentric urban region. However, here

we focus on a macro level, thus on the polycentric urban region as a network of cities.

As it can be assumed that polycentric urban regions are indeed networks (there are nodes, linkages, flows and meshes), it is likely that the same basic knowledge of synergy in networks applies to these spatial phenomena.² Depending on whether polycentric urban regions are networks of the club or the web type, the same mechanisms will lead to synergy. This means also that in polycentric urban regions, synergy is established through the mechanisms of co-operation and complementarity (and externalities involved in both).

Two important questions remain. The first is whether a polycentric urban region is a club- or a web-type network. The second is how these two mechanisms should be given a translation relevant to spatial phenomena such as polycentric urban regions. Categorising a network of cities as a club-type or web-type network seems a fruitless endeavour, given its complex nature. In fact, Capineri and Kamann (1998) state that, in real life, networks will have both club-type aspects and web-type dimensions. This is also the case with polycentric urban regions, which opens up different ways through which synergy can be established. Polycentric urban regions may be characterised as club networks when cities having similar characteristics join forces to achieve some kind of a common objective or common interests. This co-operation then generates economies of scale. Examples include co-operation between cities performing similar economic roles, such as port cities or tourist cities. But cities can also co-operate when facing similar urban problems or challenges—for instance, relating to segregation, a weak economic base, the need for efficient public transport or waste disposal. On the other hand, polycentric urban regions resemble web networks when the individual cities perform different economic roles and host complementary urban facilities, activities, residential and working environments. Comparable to the distinction between club-type networks and web-type networks, is the classification of city networks by Camagni

and Salone (1993). They refer to club-type urban networks as 'synergy networks', while web-type networks are labelled 'complementarity networks'. Although both club-type networks and web-type networks can be present in polycentric urban regions, it seems that web-type networks are of particular relevance for polycentric urban regions. With these, proximity matters in the sense that market areas overlap. Club-type networks can play a role, but they are also important in networks among distant cities (think for instance of club-type networks such as METREX or Eurocities which build on common interests).

The second question to be answered is how co-operation and complementarity (and the externalities involved) should be interpreted in the context of polycentric urban regions to provide a meaningful framework for analysis. As mentioned before, our focus is on the network at a macro level in polycentric urban regions—that is, a network between the cities rather than between firms, organisations or persons located in these cities. Consequently, co-operation is interpreted as co-operation between cities. Public administration tends to be organised in a territorial hierarchy. However, many spatial issues these days call for an approach that is formulated and implemented at multiple scales and across several administrative tiers. Additionally, an increasing number of spatial issues are, or preferably should be, addressed through a governance rather than a governmental mode. This requires the involvement of multiple public, private and organised interest-groups, thereby taking into account that different issues call for different alliances with different spatial competencies and different life-spans (see Boelens, 2000). Seen from this perspective, a focus on co-operation between cities seems too narrow. What is needed is regional organising capacity—that is, the ability to co-ordinate developments regionally through a more or less institutionalised framework of co-operation, debate, negotiation and decision-making in pursuit of regional interests in which a multitude of public and private stakeholders participate

(Meijers and Romein, 2003). The externalities that may arise depend on the utilisation and functioning of such frameworks. Synergy requires a high level of interaction which will generate the necessary network cohesion to make up for the increased interdependency (see Capello and Nijkamp, 1993). Moreover, actors must be willing and able to adjust their internal profile and external behaviour. Free-rider behaviour is to be avoided. To establish whether synergy has developed, we need to consider the extent to which such frameworks are present in polycentric urban regions.

Although some previous work on conceptualising 'complementarity' as an interurban relationship has been done (Ullmann, 1956; Lambooy, 1969; Camagni and Salone, 1993), the concept has remained rather vague, despite its increasingly frequent, but often casual, appearance in both academic writings and policy documents. This lack of conceptual clarity probably explains why the concept has so far not been empirically analysed.

In urban regions, complementarity refers to the specific nature of a relationship between two or more relatively similar activities or places. 'Activities' include economic activities, such as commercial services, or urban facilities such as education, culture and medical care. 'Places' on the other hand refer to business milieus or residential milieus. As places make up cities and most activities take place within cities, it is also expedient to refer at a macro level to cities complementing each other. For activities and places (or indirectly cities) to be complementary, they need to satisfy two important preconditions relating to supply and demand

- (1) There must be differentiation in the supply of activities and/or places.
- (2) The geographical markets of demand for these activities or places must at least partly overlap.

To give some examples, two universities are complementary if they offer different academic education, while they are at the same time recruiting their students from more or

less the same region. Similarly, two hospitals are complementary when they provide for different medical specialisations, or specialise in different kinds of treatment—for instance, standardised routine operations versus specialised knowledge-intensive care whilst serving more or less the same region. Two or more residential areas are complementary when they offer different residential milieus, thus providing alternatives to match the different preferences of a regional population (see Musterd and van Zelm, 2001). At a macro level, two cities are complementary when one specialises in, for instance, financial services and the other one in transport and logistical services, each also providing these services to businesses or citizens located in the other city.

Complementarity often leads to spatial interaction. In fact, Ullmann (1956), who describes complementarity as differentiation, argues that complementarity is the main explanation for the development of spatial interaction. Similarly, Batten (1995) states that links between the cities in a polycentric urban region (or ‘network city’ as he terms it) are forged on the basis of complementary functions rather than on the basis of distance or demand thresholds. However, mere complementarity does not suffice for spatial interaction to occur. Following Stouffer (1940), Ullmann (1956) points to the role of intervening opportunities (intervening sources of supply), as well as the role of transferability (the costs of interaction) in determining whether or not spatial interactions arise from complementarity. So, spatial interactions only partly reveal the complementarity relationships present.

One of the ideas behind the polycentric urban region concept is that it is not one city that provides a complete array of economic functions, urban facilities or residential and business environments, but rather the whole system of cities within a region. Such a situation would provide for externalities. When two cities complement each other, then the citizens and companies in one place can take advantage of the various functions the other city has to offer. These functions can then be

more specialised, as the demand market on which they build is larger given the overlapping of hinterlands. In such a way, companies, citizens and tourists can choose from a larger, more specialised and diverse collection of urban functions (public services, facilities, business services), businesses milieus and residential milieus. In other words, complementarity is strongly linked to agglomeration economies.

To sum up, synergy in polycentric urban regions is generated through

- co-operation (regional organising capacity or frameworks for co-operation and their functioning leading to horizontal synergy);
- complementarity (differentiation in the economic roles of cities, in urban facilities, in business and residential milieus coupled with a regional demand leading to vertical synergy).

In the next section, this conceptual model will be applied to the Randstad region in order to establish whether there is synergy present and developing.

4. The Randstad: Is It More than the Sum of the Parts?

The Randstad provides an excellent case for analysing the extent of synergy in a polycentric urban region, not only because many authors have claimed that it is a classic stereotype of such a network of cities (see for instance Stein, 1964; Hohenberg and Lees, 1985; Batten, 1995), but also because several authors have suggested the presence of synergy. For instance, Camagni and Salone (1993) refer to the Randstad as an example of a complementary network made up of specialised and complementary centres. This is fully endorsed by van der Knaap (1994) who states that in the Dutch context there seems to be a dominance of complementary networks, based on vertical integration, spatial specialisations and complementary urban centres. This would be particularly exemplified in the Randstad, where it is mainly expressed in the relations between the larger and smaller cities. After

briefly introducing the Randstad region, we will analyse whether such claims can be based on empirical evidence.

The Randstad is the horseshoe-shaped urban constellation in the western part of the Netherlands (see Figure 1). The Dutch word 'Rand' means 'rim' and refers to the position of the Randstad encircling a green open area called the Green Heart. Seven million people live in the Randstad (44 per cent of the Dutch population) and 45 per cent of the national employment is located in what is less than 20 per cent of Dutch territory. The anchors of

the Randstad are formed by the four largest cities of the Netherlands: Amsterdam, Rotterdam, The Hague and Utrecht. Together with some dozens of other medium-sized and smaller cities and the absence of one predominant centre, the Randstad can be characterised as a stereotypical polycentric urban region. This polycentric pattern is basically inherited from the past, as fragmented political and administrative structures have prevailed in this area (in fact, in a large part of north-west Europe) for centuries (Dieleman and Faludi, 1998b). However, the Randstad, as it

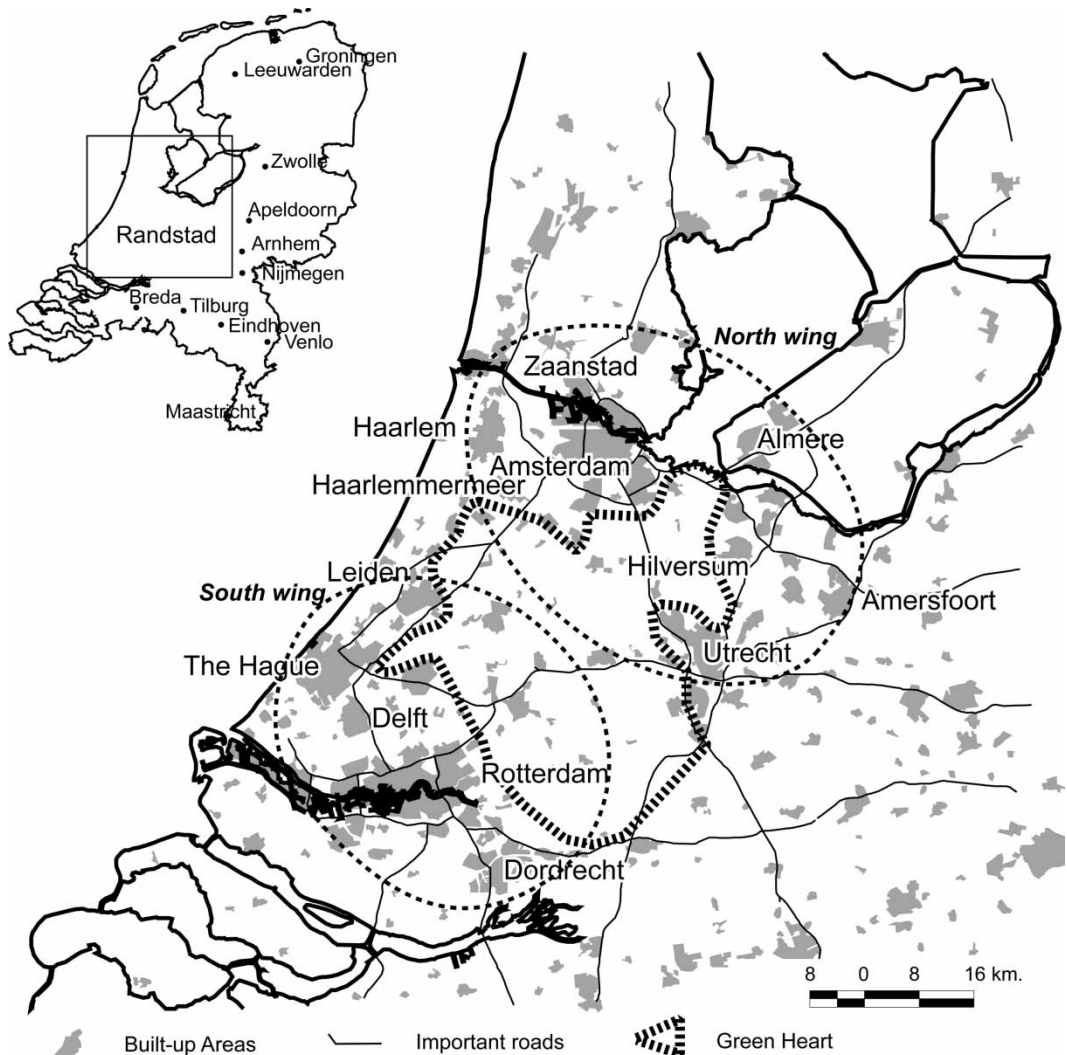


Figure 1. The Randstad.

is nowadays, is also the result of urban and regional planning, being, with its counterpart the Green Heart, at the core of Dutch planning policies since the 1950s (Zonneveld, 1991; Faludi and van der Valk, 1994). A common division of the Randstad is into a north wing (including Amsterdam, Utrecht and surrounding cities) and a south wing (The Hague, Rotterdam and surrounding cities). Suggestions have been made that these two wings provide a more accurate definition of coherent urban regions than does the Randstad (van der Laan, 1998; Kloosterman and Lambregts, 2001).

The Randstad region still dominates in Dutch planning. The proposals for a new national spatial policy introduce the concept of 'urban networks' and the Randstad is the most important one (Ministerie van Volkshuisvesting, Ruimtelijke Ordening en Milieubeheer, 2004). In order to increase the Randstad's competitiveness, emphasis is put on the need for synergy. This pursuit of synergy is also expressed by several other networks of relevant actors in the Randstad. For instance, the Delta Metropolis Association (see also next section) states as the first guiding principle for the development of the Randstad

In a European perspective, the development of the Delta Metropolis is aimed at the achievement of synergy in the activities of the people living there; more synergy is possible and needed in the light of social, economic and cultural activities and efforts aimed at a sustainable environment (Delta Metropolis Association, 2000, p. 1).³

And, the 'Projectgroep Randstadinbreng Vijfde Nota', in which several provinces, regional authorities and the major cities in the Randstad were represented, has presented as a first major administrative outline that

The urbanised western part of our country needs a quality leap towards a Blue Green Delta Metropolis. This must be seen in the context of the pursuit to let the Randstad be more than the sum of the parts, in order to take a competitive position in

Europe. Co-operation and fine-tuning between networked cities, each having their own specialities, and the wetlands of the Green Heart will give the Randstad as a whole added value and will increase the quality of life. The necessary increase in quality requires an extended support for facilities and amenities, distinct profiling of areas and an increase in diversity (Bureau Regio Randstad, 2001, p. 6).

The following sections present the results of an explorative analysis of synergy in the Randstad, addressing co-operation and complementarity. Most attention, however, will be paid to the synergy mechanism of complementarity for two reasons. First, it was argued that, compared with co-operation, this mechanism is most prominent in polycentric urban regions as proximity matters. Secondly, claims for the Randstad being a complementary network have been made (Camagni and Salone, 1993; van der Knaap, 1994).

5. Co-operation or Horizontal Synergy in the Randstad

There are three administrative tiers in the Netherlands: national government, provinces and municipalities. The Randstad extends over four different provinces, which complicates the recognition of and dealing with the complex, multiscale interplay of spatial trends and forces in an integrated way. Although several attempts have been made to add a formal administrative tier, in particular at the supralocal scale, the existing framework has proved to be rather resistant to changes. It has become slowly apparent that multilevel governance requires co-operation across scales and across actors, including private actors.

In the Randstad, it seems that actors are increasingly aware of this need. A large variety of co-operation networks has emerged in this region in an attempt to overcome the lack of administrative layers between the municipal and provincial levels and between the provincial and national levels. Around the 4 largest Randstad cities,

city-regions have been formed including the large cities and 10–20 adjacent municipalities. These co-operation platforms address issues such as transport, traffic, regional spatial development, housing, employment, economic affairs and youth welfare. Several more or less similar co-operation networks exist around smaller cities (Dordrecht, Leiden) or more or less homogeneous regions in terms of economic activities or location. At a higher spatial level, we find co-operation networks in the northern and southern wings of the Randstad. Compared with the north wing, the co-operation in the south wing is closer, probably due to a greater urgency for co-operation because of a lower level of economic development. The co-operation in the south wing involves the municipalities of The Hague and Rotterdam, the province of South Holland and various co-operation platforms at the city-region level. Finally, also at the scale of the entire Randstad, two interesting co-operation networks have emerged in recent years. The first is a formal co-operation between the four provinces, four regional authorities (city-regions) and four major cities in the Randstad, together organising the 'Bureau Regio Randstad' (Randstad Agency). Their objectives are to foster a balanced and dynamic development of the western part of the Netherlands, the Randstad in particular, as an entity consisting of high-quality urban and rural environments and to strengthen the international competitiveness of the Randstad, in particular within Europe. Next to this formal co-operation between public actors, an informal co-operation platform has emerged: the Delta Metropolis Association. It was established in 2000 by 12 municipalities and 4 chambers of commerce from within the Randstad. Being an open network, the number of members has gradually grown and now also includes housing corporations, organisations of the agriculture and horticulture branches, an employer's organisation, the transport sector, environmental organisations and water boards. Negotiations are taking place to involve other organisations, for instance the universities, as well.

The co-operation platform functions as a think-tank predominantly occupied with the central ambition to let the 'in principle yet present metropolis grow to full stature' (article 2 of the statutes of the Association). All in all, it can be typified as a lobby group for the interests of the Randstad region. Interestingly, all these co-operation networks were initiated by actors within the region, rather than dictated by a higher level of government.

The patchwork of co-operation networks in the Randstad provides for a considerable regional organising capacity in the Randstad. All these networks are examples of club networks as they share a common objective and have at least partly corresponding interests and together fund the organisation of the network. Thus synergy is likely to be achieved, although the development of externalities depends on the functioning of these often rather new co-operation networks. Interestingly, Capello (2000) has demonstrated the development of externalities for actors when participating in a co-operation network in an active and serious way. It needs to be said that this co-operation does not mean that the cities do not compete with each other. On the contrary, they have competed for centuries and will continue to contend in the future. However, the awareness that many issues call for a co-ordinated approach coupled with the idea that regions have become the principal geographical platform for competition (see for instance Storper, 1995; Scott, 1998) fuelled a reconsideration of the intra-regional interdependencies resulting in more co-operation. Interesting in this respect is that, at the level of the Randstad or its two wings, co-operation networks are formed between the larger cities, while the main competitors of these cities tend to be the nearby smaller (satellite) cities, who have profited from the process of selective out-migration of middle-class and higher-income households as well as economic activities (van der Wouden and de Bruijne, 2001). Obviously, the co-operation networks in the Randstad still have to prove themselves, but some successes are already visible—for instance, the Randstad region office in

Brussels which takes care of the region's interests in the European context, and the joint approach towards policy proposals, investment plans, etc. emanating from the national government. Perhaps the biggest success so far is that the Randstad region features prominently in spatial as well as in many sectoral national policies. Obviously, there is more horizontal synergy in the Randstad than there was before, even though this horizontal synergy is hard to quantify at this moment.

6. Complementarity or Vertical Synergy in the Randstad

The presence of complementarity in the Randstad has already been assumed by several authors (Camagni and Salone, 1993; van der Knaap, 1994), although these assumptions lack strong empirical justification. We have seen that complementarity relates to differentiation in economic roles, urban facilities and business and residential environments. Moreover, a geographical overlapping of demand markets for these activities and places is also important. Consequently, analysing such a multifaceted concept as complementarity is a complex undertaking. Here, the exploratory analysis of the presence of complementarity in the Randstad is limited to one of these facets—namely, the differentiation in economic roles, or economic profiles, of the main cities within the Randstad. Differentiation in economic roles also indicates to some extent the differentiation in business environments, as one city may provide better conditions for certain firms and this then becomes explicit in its economic profile. Economic profiles also present an indication of differentiation in urban facilities as these are part of a city's economic profile. Another reason for focusing on the macro level of economic roles of cities is that the extent of complementarity that is generally assumed to be present in the Randstad is usually linked to the division of labour between its main cities (see for example, van der Wouden and de Bruijne, 2001). Of course, the need for selectivity here does not deny the fact that the issue of

complementarity requires extended analysis and, consequently, this limits the significance of the results as regards the synergy derived from complementarity in the Randstad.

The economic profile of a city is made up by the number of jobs in that city classified by economic sector. For our analysis, we used a data-set derived from the National Information System on Employment (LISA) database. This database is a registration of all the establishments in the Netherlands, including government and non-commercial organisations. An establishment in this database is defined as a location of a firm, organisation, institution or independent profession in which or from which an economic activity or independent liberal profession is being practised by at least one employed person. Multi-establishment firms have separate recordings for each establishment. The economic activities are coded according to the European Union wide NACE Rev. 1 classification (*Nomenclature statistique des Activités économiques dans la Communauté Européenne*), which is, up to two digits, similar to the ISIC Rev. 3.1 (International Standard Industrial Classification of all Economic Activities). The database has been updated annually since 1996. Our data-set contains the 1996 and 2002 data on the number of jobs classified according to the two-digit NACE Rev. 1 classification of economic activities for the municipalities of Amsterdam, Rotterdam, The Hague and Utrecht. For reasons of comparison, we also have data-sets for the main cities in the north wing (municipalities of Amsterdam, Haarlem, Zaanstad, Utrecht, Amersfoort, Almere, Haarlemmermeer and Hilversum) and south wing (Rotterdam, The Hague, Leiden, Dordrecht, Zoetermeer and Delft).

In this paper, correspondence analysis is used to analyse the differentiation in the economic roles of cities. Correspondence analysis is a technique to analyse the association between rows and columns of a table or matrix by representing the rows and columns as points in a low-dimensional Euclidean space (in practice, often a two-dimensional plot). Categories with similar distributions

will be represented as points that are close in space and categories that have very dissimilar distributions will be positioned far apart. For an extensive discussion of correspondence analysis, see Greenacre (1993) and Clausen (1998).⁴ Although often used as a tool to enable graphic interpretation of complex data, correspondence analysis also provides a single statistic that describes the extent of differentiation in the economic profiles of a group of cities. This statistic is called the total inertia. Total inertia is a measure of the extent to which the profile points are spread around a centroid, representing the average profile. The larger the distance of the category points from the centroid, the higher the inertia. The highest attainable inertia is equal to the dimensionality of the problem (in our case, the number of cities—1). This maximum would be reached if all the cities have completely different economic profiles, whereas zero inertia is attained when they all have exactly the same economic profiles. In reality, values will be far from the maximum, as reaching the maximum value would imply—for example, that all schools are located in one city, all supermarkets in another one and all doctors in yet another one. In other words, cities have a large component of employment in non-tradeable economic activities. Correspondence analysis corrects for such activities in the sense that they do not, or hardly, contribute to the total inertia statistic. In order to enable a comparison of the inertia between polycentric urban regions, we defined a complementarity ratio for which we normalised the total inertia by

dividing it by the maximum total inertia possible and multiplying this by 100, resulting in a value between 0 and 100.

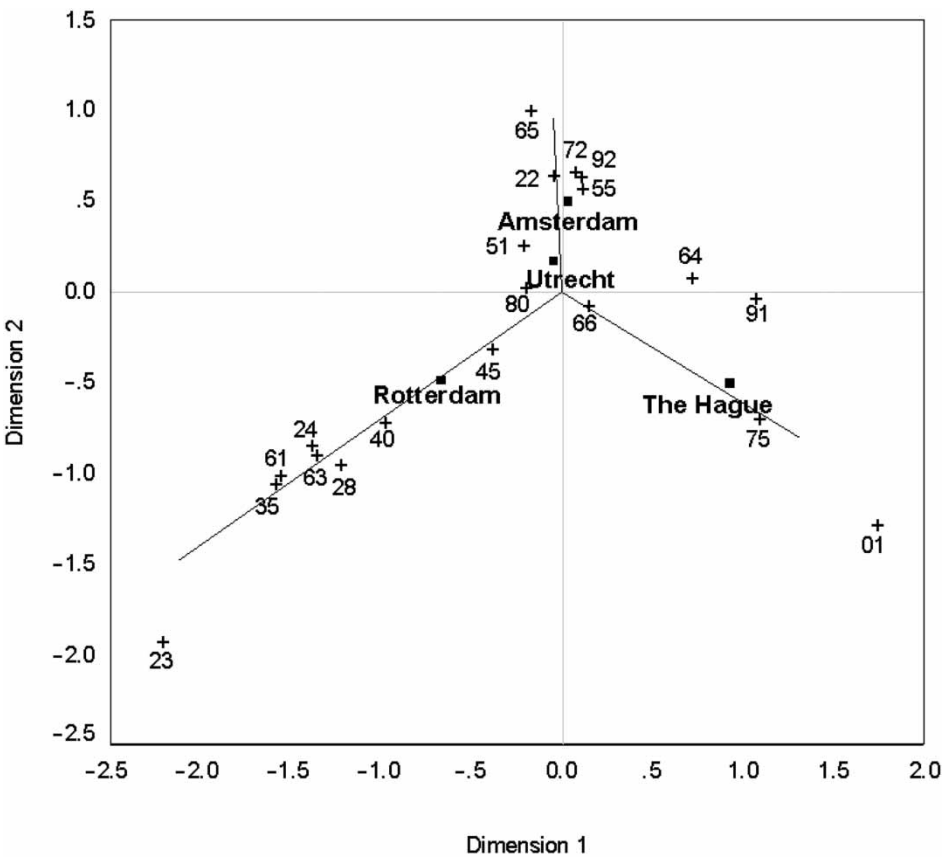
Table 1 presents the results of our analysis, the total inertia and complementarity ratios for the Randstad and its south and north wings for 1996 and 2002. The table reveals clearly that the economic profiles of cities within the Randstad, or its south and north wings, are becoming less differentiated. This trend towards more homogeneous economic roles for cities indicates a decreasing complementarity in terms of economic roles (on average, a decrease of nearly 10 per cent in 6 years). The north wing of the Randstad is the most complementary region, with a ratio of 6.3 in 2002. This is 29 per cent higher than in its counterpart, the south wing, and the Randstad as a whole when defined by the 4 cities. This is 66 per cent higher than if the Randstad is defined using all the larger cities. However, the ratio in the north wing is decreasing faster than in the south wing, but not as fast as in the Randstad defined by the 4 largest cities.

One of the main advantages of correspondence analysis is that it graphically displays associations, thus enabling an easier interpretation of complex contingency tables. Here, we are interested in associating cities with economic activities (see Figure 2). The 2 dimensions displayed represent a reasonable 84.2 per cent of the total inertia. The two axes together indicate the origin (0.0), which resembles the average profile of the 4 cities. This plot requires careful interpretation. If 2 cities lie close together, then their economic

Table 1. Total inertia and complementarity ratios in the Randstad and spatial sub-divisions in 1996 and 2002

Polycentric urban region	Total inertia		Complementarity ratio		Trend (percentage)
	1996	2002	1996	2002	
Randstad ($N = 4$)	0.170	0.148	5.7	4.9	–12.9
Randstad ($N = 14$)	0.545	0.493	4.2	3.8	–9.5
North wing ($N = 8$)	0.489	0.440	7.0	6.3	–10.0
South wing ($N = 6$)	0.263	0.245	5.3	4.9	–6.8

N = Number of cities included.



Key

01	Agriculture, hunting and related service activities	61	Water transport
22	Publishing, printing and reproduction of recorded media	63	Supporting and auxiliary transport activities; travel agencies
23	Manufacture of coke, refined petroleum products and nuclear fuel	64	Post and telecommunications
24	Manufacture of chemicals and chemical products	65	Financial intermediation, except insurance and pension funding
28	Manufacture of fabricated metal products	66	Insurance and pension funding
35	Manufacture of other transport equipment	72	Computer and related activities
40	Electricity, gas, steam and hot water supply	75	Public administration and defence; compulsory social security
45	Construction	80	Education
51	Wholesale trade and commission trade	91	Activities of membership organisations n.e.c.
55	Hotels and restaurants	92	Recreational, cultural and sporting activities

Figure 2. Cities associated with economic activities in the Randstad ($N = 4$), 2002 (percentage of inertia explained: 84.2 per cent).

profiles are more or less similar. The same condition applies to the economic activities. Economic activities lying close together are more or less similarly distributed between the cities. The distances between cities and economic activities are more complicated, since these are not defined as chi-squared distances. All cities influence the location of an economic activity and, the other way around, all economic activities contribute to the location of a city. In general, cities and activities will be close to each other when the observed value for this pair of points in the table is larger than expected, and the distance will be large when the observed value is less than the expected value. For reasons of clarity, only those economic activities that contribute to at least 1 per cent of the total inertia value are depicted. Economic activities that do not meet this threshold value are either insignificant in terms of the number of jobs or because the distribution of jobs in this activity over the 4 cities is similar to the distribution of all jobs over these cities. This is for instance the case with retail trade, where the expected number of jobs in the 4 cities is more or less equal to the observed number.

In 2002, the three largest cities, Amsterdam, Rotterdam and The Hague seem to have relatively distinct profiles, while Utrecht has a more general and average profile and is, therefore, located closer to the origin. Still, Utrecht's economic profile is more similar to that of Amsterdam than to the other cities. Out of the origin it is possible to distinguish three axes in which a number of economic activities are grouped that are dominated by one of the three main cities. Amsterdam has a relatively dominant position in the commercial services sector, in particular in financial intermediation, Internet and communication technology, publishing and printing, recreation, culture and sports, as well as hotels and restaurants. Rotterdam, on the other hand, holds a strong position in manufacturing and transport, undoubtedly related to its large port. This includes heavy industries such as the petrochemical and chemical industries, the manufacturing of fabricated metal products and

of transport equipment. Not surprisingly, Rotterdam specialises in water transport and support and auxiliary transport activities. Finally, Rotterdam has relatively more jobs in the construction and public utilities sectors. The Hague, the seat of government, is particularly dominant in public administration. Moreover, it has a position in agriculture, but this seems predominantly due to the presence of a number of agricultural interest-groups. The Hague shares, with Amsterdam and Utrecht, a strong position in activities of membership organisations and post and telecommunications. Utrecht is relatively more associated with wholesale trade and education. In general, it seems that there is a considerable division of labour between the three largest cities in the Randstad, each specialising in either commercial services (Amsterdam), manufacturing and transport (Rotterdam) or public administration (The Hague). The spread of economic activities over the four cities is to a large extent similar to the pattern observed for 1996.

Figure 3 presents the developments in the economic profile of the four main cities in the Randstad between 1996 and 2002. The origin here is the weighted average of the average economic profiles in 1996 and 2002. Figure 3 provides an accurate picture of developments in the differentiation in economic profiles (roles) of the four main cities in the Randstad. Rotterdam accounts for the largest change in position as it became considerably less different in economic profile given its move towards the average economic profile. This could be due to the number of jobs in the manufacturing and transport sectors decreasing relative to other sectors. Alternatively, other cities may have seen a considerable increase in the number of jobs in economic activities associated with them. This is for instance the case for The Hague which, contrary to Rotterdam, moved further away from the origin, indicating increased differentiation. The number of jobs in the public administration sector increased considerably in the 1996–2002 period. Similar to The Hague, Utrecht's economic role became more distinct, although trends

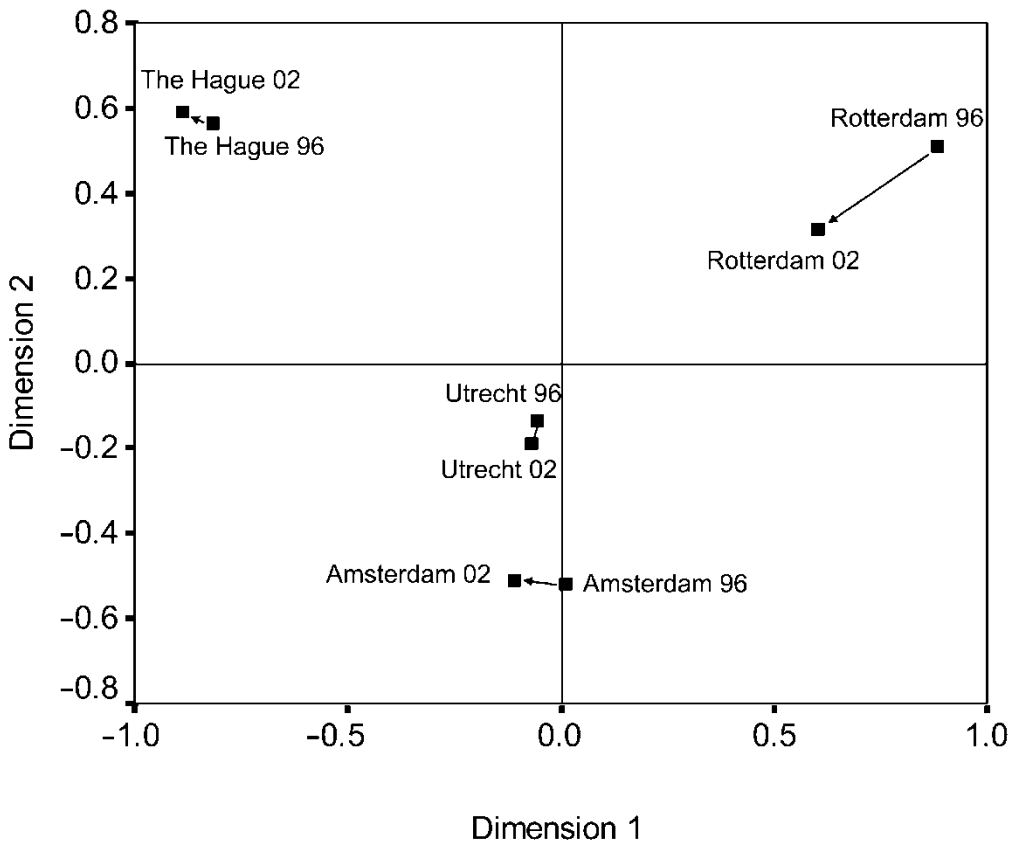


Figure 3. Development in differentiation of economic roles of Randstad cities between 1996 and 2002 (percentage of inertia explained: 79.4 per cent).

indicate that it will converge with that of Amsterdam. Amsterdam's economic role remained more or less equally distinct. The overall conclusions regarding synergy in the Randstad will be presented in the conclusions.

7. Conclusions

The network metaphor first entered social sciences several decades ago. Recently, it has also proved to be a fruitful inspiration base for exploring the relationships between actors, phenomena and elements in the spatial sciences. Thus it may be considered the contemporary equivalent to the systems approach which, although still relevant, had its heyday in the 1970s and also directed research in a wide variety of fields of study. In urban and regional planning, the network

metaphor has been increasingly applied to describe interurban relationships. Analysing the synergy concept in economic network theory and transferring the findings to spatial phenomena such as polycentric urban regions reveal that it is exactly these relationships between cities that determine whether or not synergy is present. In particular, it is the nature of these interurban relationships that is decisive when considering synergy. It was found that polycentric urban regions can indeed be more than the sum of their parts by means of co-operative and, in particular, complementary relationships and externalities arising from them.

The results of our explorative analysis of synergy in the Randstad are mixed. The first synergy mechanism, co-operation, seems to be increasingly present in the Randstad. The

bottom–up establishment of both formal and informal frameworks for a regional co-ordination of regional issues and developments has led to more regional organising capacity than there was before. The wide variety of co-operating actors present in the two frameworks operating at the scale of the Randstad are even explicitly aiming for increased synergy. Although most partnerships at the moment do not yet have an extensive track record of successes, given their relatively recent establishment, it appears that these club-type networks provide much better conditions for achieving horizontal synergy than ever before. Thus, it is likely that synergy has increased.

We argued that complementarity, the second synergy mechanism, is of particular relevance for polycentric urban regions. Our analysis here was limited to differentiation in the economic role performed by cities, which we inferred from the economic profile of cities within polycentric urban regions. The analysis revealed that the main cities in the Randstad perform distinct roles, each of them specialising in either commercial services, manufacturing and transport, public administration, or trade and education. At the same time, the extent of complementarity in economic roles diminished during the 1996–2002 period by almost 13 per cent, leading to less synergy at the macro level of the cities in the Randstad. Following the common divide of the Randstad into a north wing and a south wing reveals that complementarity in the north wing is substantially higher than in the south wing. Whether or not the balance between both synergy mechanisms is shifting to more or less synergy in the Randstad remains a crucial question. Answering it requires further research into other facets of complementarity at the micro level of certain activities and particular places, the extent to which differentiation in these activities and places is matched by a regional demand for them, as well as of the externalities present in the co-operation frameworks.

The decline in complementarity in terms of economic profiles of cities should be

looked at in the proper perspective. Other research on developments in the sectoral composition of cities has shown that the sectoral specialisation of cities of all sizes has already been declining considerably for a long time (Duranton and Puga, 2001). From this viewpoint, our findings regarding the development in synergy through the mechanism of complementarity are perhaps not surprising, nor as negative as they may seem. Interestingly, Duranton and Puga (2001) argue that cities are increasingly distinguished by their functional specialisation rather than by their sectoral specialisation. They demonstrate that, in US cities, headquarters and business services functions of manufacturing companies cluster in larger cities, whereas production functions cluster in smaller cities. Similar processes are likely to be found in the Randstad. Although the spread of sectoral economic activities over the region is becoming more homogeneous, it may well be that, within a sectoral economic activity, different functions (for instance, front-offices versus back-offices, headquarter versus production, high-skill top-level functions versus low-skill routine functions, etc.) can be found in different cities.

Finally, we need to draw a link with the contemporary and emerging debate on the changing spatial organisation of urbanised regions and in particular on the changing relations between cities. Whereas interurban relationships have long been defined in terms of hierarchy, it is increasingly assumed that these vertical Christaller-like type relationships are being replaced, or at least supplemented with more horizontal network-like relationships between cities such as complementarity and, at the institutional level, co-operation. This new model of spatial organisation is generally referred to as a network urban structure. From our analysis, it follows that the development of a network urban structure in urban regions is beneficial in terms of synergy as the horizontal network-like relationships between cities that build such a structure correspond exactly with the synergy mechanisms.

Notes

1. The term 'city networks' is generally applied to: systems of distant cities linked to each other by some functional characteristic; and, a system of proximal cities more or less located within each others' functional hinterlands. Polycentric urban regions are city networks in the latter meaning.
2. A comparable analogy between firm networks and urban networks is for instance drawn by Dematteis (1991), Emanuel (1990) and Camagni (1993).
3. 'Delta Metropolis' is synonymous with 'Randstad'.
4. See Kloosterman and Lambregts (2001) for an earlier example of the application of correspondence analysis. Rather than analysing general economic profiles of cities, they focus on patterns of convergence in business start-up profiles of cities in the Randstad.

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